2025年度(令和7年度)

前期日程

英語 (80分)

注意事項

1. 試験開始の合図があるまで，この問題冊子を開いてはいけません。
2. 問題冊子は，1ページから17ページまであります。解答用紙は，2枚からなっており，「1」はマーク式，「2」は記述式です。ページの脱落等に気付いたときは，手をあげて監督者に知らせなさい。
3. 監督者の指示に従って，すべての解答用紙の受験番号欄に受験番号を記入し，マーク欄にマークしなさい。
4. 解答はすべて，解答用紙の指定された欄にマーク又は記入しなさい。
5. 受験番号のマーク時及び解答用紙「1」の解答には，必ず黒鉛筆(H，F，HBに限る)及びプラスチック製の消しゴムを使用し，「マーク例」に従って，正しくマークしなさい。黒鉛筆以外のもの(シャープペンシル等)を使用してマークした場合，マークが薄い場合，一部分しかマークしていない場合，訂正箇所を消しゴムできれいに消していない(消し跡が残っている)場合は，マークが正しく読み取れず，採点できないことがあります。
6. 解答用紙の網掛け部分及び※を付した欄には，何も記入してはいけません。
7. 問題冊子の白紙と余白は下書きに適宜利用してもよいが，どのページも切り離してはいけません。
8. 試験終了後，この問題冊子は持ち帰りなさい。

I

1 Read the following text and answer the question below.

Everyone knows that you should brush your teeth at least twice a day: once in the morning and once before bedtime. But when it comes to your morning routine, is it better to brush before or after breakfast? The truth is that few studies have looked into this question and their results have been mixed and limited, so there isn't a definitive answer, said Dr. Apoena de Aguiar Ribeiro, a pediatric dentist and microbiologist at the University of North Carolina at Chapel Hill. But, she and other dentists said, there are pros and cons for each option to keep in mind. Ultimately, the decision boils down to what works for you, Dr. de Aguiar Ribeiro said, because the science is sparse and "there is no consensus in the literature." Pediatric dentist Dr. Quinonez agreed: There is no proof either way, she said, so all of the arguments are largely theoretical. "It's highly debated, and it ultimately may not make a significant clinical difference," she said. In the end, Dr. Quinonez said, the most important question to ask yourself is this: What brushing habit will make me most likely to brush my teeth consistently? Once you have your answer, she said, work it into your routine and stick with it.

【出典(一部改変): https://www.nytimes.com/】

Which of the following is the main idea of the text?

A. Few researchers have studied whether we should brush our teeth before or after breakfast.

B. Some people believe that it's better to brush your teeth after breakfast.

C. We should brush our teeth at least twice a day.

D. We should make morning brushing a habit, whenever it is.

2 **Read the following text and answer the question below.**

A giant 3D printer, which is big enough to make a house, has been developed at the University of Maine. The university says it has beaten its own record for the world's largest polymer 3D printer — with the new printer four times bigger than the previous machine. The Factory of the Future 1.0 (FoF 1.0) can print objects 29 meters long — approximately the length of a blue whale.

But experts say, despite the breakthrough, most of us will continue to live in homes made of bricks and mortar. Large 3D printers, such as that presented in Maine, are more likely to be used to make parts of houses rather than full homes, says Dr. Eujin Pei, who is an expert in additive manufacturing at Brunel University. He says they can be used for historical restoration, "not to print a house, but sometimes to print decorative parts, or sometimes to restore some parts." They may become useful after natural disasters to manufacture parts of destroyed buildings, he told the BBC.

The University of Maine says it hopes the printer can be used to make affordable housing, as well as bridges, boats, and wind turbines. It can print up to 227 kilograms of material per hour. While 3D printers often print using plastic, the university hopes to include more sustainable materials and prioritize recycled polymers.

【出典(一部改変): https://www.bbc.com/】

Which of the following statements is true?

A. A giant 3D printer revealed at the University of Maine can print objects 29 meters long, which is four times longer than a blue whale.

B. FoF 1.0 is made of recycled polymers, and people can afford to buy it.

C. FoF 1.0 may be useful for the partial repair of historical architecture or damaged buildings rather than for making full houses.

D. The invention of FoF 1.0 is likely to make most people get rid of brick and mortar houses.

3 Read the following text and answer the question below.

Ludwig van Beethoven's Ninth Symphony was first performed exactly 200 years ago and has since become probably the work most likely to be used for political purposes. It was played at the 1936 Olympic Games in Berlin. It was performed in that city again on Christmas 1989 after the fall of the Berlin Wall, when Leonard Bernstein replaced the word "Joy" in the choral finale with "Freedom".

Beethoven might have been surprised at the political attraction of his masterpiece. He was interested in politics, but only because he was deeply interested in humanity. He was concerned with moral behavior and the larger questions of right and wrong affecting all of society. Especially significant for him was [X], which he associated with the rights and responsibilities of the individual. The closest he comes to a political statement in the Ninth Symphony is a sentence at the heart of the last movement, in which voices were heard for the first time in a symphony: "All men become brothers."

Music, if you study it properly, is a lesson for life. There is much we can learn from Beethoven. He is the master of bringing emotion and intellect together.

【出典(一部改変): https://www.nytimes.com/】

Which of the following best fills [X]?

A. freedom of thought and of personal expression

B. his reputation compared to other composers

C. integration of new musical techniques

D. receiving payment for his work

4 Choose the best answer from A—D to rearrange the following sentences in the best order.

① Fingerprints form before birth. A person's fingerprints may spread out slightly over the course of their life. But the overall pattern stays the same.

② However, criminals have attempted to erase these prints, complicating investigations. What is more, it has been hard to tell whether fingerprints found at a crime scene were left by the criminal or someone else at another time.

③ Nearly everyone in the world carries a unique ID right at their fingertips — their fingerprints. That is why some phones can be unlocked with a user's fingertip. And why fingerprints left at a crime scene can help identify the criminal.

④ One team has found a way to uncover fingerprints that have been wiped down. The technique relies on the fact that chemicals in a person's fingerprints corrode metal surfaces. That corrosion can be detected even if a print is not visible.

⑤ Other scientists have found a way to estimate the age of a fingerprint. This can expose whether the fingerprint dates to the time of a crime or before or after.

⑥ When someone touches an object, they may leave a stamp of their fingerprint on it. This imprint may be made of oil, dirt, or other substances that were on the person's skin. Forensic scientists have long gathered such fingerprints from crime scenes. Those prints can offer clues about who has been at a certain place and what they did there.

【出典(一部改変): https://www.snexplores.org/】

A. ③—①—⑥—②—④—⑤
B. ③—⑥—④—⑤—②—①
C. ⑥—②—③—④—⑤—①
D. ⑥—④—⑤—③—①—②

II

Read the following text and choose the most appropriate word or phrase from A—D **to fill in** (1)—(10).

We all know someone who never seems to be on time, whether it's to a lunch date or a work meeting. But is there a good explanation for why some people are always late? The habit of being tardy probably results from a number of factors, including time perception, time management, and personality, experts say.

"It is likely that there's a mechanism in the brain that causes some people to be late for meetings (1) they guess wrongly about the time it will take them to get there," Hugo Spiers, a professor of cognitive neuroscience at University College London and the co-author of a 2017 study said. The hippocampus is a region of the brain which processes some aspects of time, such as (2) when to do something and how long it takes. Research suggests that neurons in the hippocampus acting as "time cells" contribute to our perception and memory of events, but why exactly some people perpetually underestimate time is unclear.

One factor may be how familiar we are with a space. For the 2017 study, Spiers asked 20 students who had newly moved to London (3) sketch a map of their college district and estimate travel times to different destinations. While the students' space estimates expanded if they knew an area well, their estimate of travel time (4) with familiarity. "If you're very familiar with a space, you start to discount the hassle it will take," Spiers said. In some cases, people who are late may not allow enough time to complete tasks unrelated to travel, such as getting ready in the morning. Research suggests that we make time estimates based on how long we think tasks (5) us in the past, but our memories and perceptions aren't always accurate.

"If we have a lot of experience performing a task, we are more likely to

underestimate how long it will take," Emily Waldum, a researcher at Campbell University and the lead author of a 2016 study said. In the study, Waldum found that environmental factors, such as music, can (6) your sense of time. Specifically, Waldum showed that when doing a general knowledge questions task, some people incorrectly estimated the task's length based on the number of songs they heard playing in the background. Younger adults tended to inflate their time estimates if they heard four short songs compared with two longer songs, something that didn't seem to (7) older adults' perception of time.

Another environmental factor may be crowdedness. In a 2022 study, researchers asked participants to estimate the length of more or less crowded simulated subway trips. They found that crowded commutes felt like they took 10% longer than less busy rides, which was (8) to it being an unpleasant experience.

Other research has shown that people juggling several tasks at once are less likely to remember and complete other scheduled tasks on time. "The best plans can (9) simply because we don't have enough attentional resources left to carry them out successfully," Waldum said.

Latecomers sometimes don't perceive themselves as such. That's because people who run behind schedule tell themselves and others that they can be punctual. We can be on time when it (10), when there will be negative consequences for our lateness, like missing a flight.

【出典(一部改変)：https://www.livescience.com/】

(1)	A. because	B. before	C. though	D. unless
(2)	A. building	B. consuming	C. remembering	D. reversing
(3)	A. and	B. helped	C. to	D. would
(4)	A. decreased	B. enhanced	C. increased	D. suspended
(5)	A. are taking	B. have taken	C. take	D. would take
(6)	A. borrow	B. confuse	C. demand	D. force
(7)	A. increase	B. indicate	C. indirect	D. influence
(8)	A. aimed	B. happened	C. linked	D. proved
(9)	A. avoid	B. fail	C. start	D. work
(10)	A. depends	B. matters	C. means	D. seems

Ⅲ

Read the following text and answer the questions below.

Recently, a group of 60 prominent scientists and engineers met behind closed doors at the University of Colorado Boulder. Their agenda: Mars colonization. The goal of this "Mars workshop" was to begin making concrete plans for landing, building, and maintaining a human colony on Mars within the next 40 to 100 years.

This workshop signals the growing momentum and reality behind plans to actually send humans to Mars. But while members of this committee ask whether we could live there, others still ask whether we should.

We have known for some time that the journey to Mars for humans would be hard. It's expensive. It's dangerous. It's boring. Humans should leave the red planet alone.

It is hard to forget the images of Elon Musk's dark red Tesla e-vehicle floating through space. Launched atop the Falcon Heavy rocket, Musk hoped to shoot the Tesla into orbit with Mars. A silly ① stunt, for sure — but also a marvelous display of technical competence.

But not everyone was happy. Unlike every previous craft sent to Mars, this car had not been sterilized to kill bacteria. And for this reason, some scientists described it as the "largest load of earthly bacteria to ever enter space."

As it happens, the Tesla missed its orbit. At the time of writing, it is 142 million kilometers from Mars, drifting through the darkness of space. But the ② episode illustrates the first argument against human travel to Mars: contamination.

If humans did eventually land on Mars, they would not arrive alone. They would carry with them their earthly microbes. Trillions of tiny life forms.

There is a real risk that some of these microbes could find their way onto

the surface of Mars and, in doing so, confuse — perhaps irreversibly so — the search for Martian life. This is because we wouldn't be able to distinguish indigenous life from the microbes we'd brought with us. Our presence on Mars could jeopardize ③ one of our main reasons for being there — the search for life.

Furthermore, there is no one way of knowing how our microbes may react with the vulnerable Martian ecosystem. In *Cosmos*, the late Carl Sagan wrote, "If there is life on Mars, I believe we should do nothing with Mars. Mars belongs to the Martians, even if the Martians are only microbes, the preservation of that life must, I think, be considered more important than any other possible use of Mars."

Of course, one easy way to minimize the risk of contamination is to send robots to Mars instead of humans — the second argument against a manned trip to Mars.

Robots have several inherent ④ advantages. They are much cheaper than humans because they don't require a vast support infrastructure to provide things like water, food, and breathable air. They are immune to the risks of cosmic radiation and other dangers involved in space travel. And they won't get bored.

With the rapid pace of progress in robotics and AI, it is likely that the effectiveness of these non-human explorers will only increase. Robots on Mars will be able to carry out increasingly complex scientific research, accessing craters and canyons that humans might find too difficult to reach — and perhaps even drilling for Martian microbes.

The most polarizing ⑤ issue in the Mars debate is arguably the tension between those dreaming of a second home and those prioritizing the one we have now.

Faced with a growing list of threats — climate change, overpopulation, nuclear war — some scientists believe that we have reached "the point of no return" and have no choice as a species but to become multi-planetary —

starting with the colonization of Mars.

However, not everyone agrees. A majority of US adults believe that NASA's number one priority should be fixing problems on Earth. The billions — if not trillions — of dollars needed to colonize Mars could, for example, be better spent investing in renewable forms of energy to address climate change or strengthening our planetary defenses against asteroid collisions.

And of course, if we have not figured out how to deal with problems of our own making here on Earth, there is no guarantee that the same fate would not befall Mars colonists.

Furthermore, if something truly horrible were to happen on Earth, it's not clear Mars would actually be an effective salvation. Giant underground shelters on Earth, for example, could protect more people, more easily than a colony on Mars.

And in the event of an apocalyptic scenario, it is possible that the conditions on Earth — however terrible — may still be more hospitable than the Martian wasteland. Let's not forget that Mars has next to no atmosphere, only one third gravity and is exposed to surface radiation approximately 100 times greater than on Earth.

The arguments above show that we are perhaps not ready to go to Mars — at least, not today.

【出典(一部改変): https://www.theguardian.com/】

1 Choose the word from A—D which is closest in meaning to ①—⑤.

① A. demonstration B. disaster
C. escape D. station

② A. crime B. damage
C. event D. indication

③ A. empower B. endanger
C. ensure D. guarantee

④ A. artificial B. built-in
C. complex D. unknown

⑤ A. controversial B. convenient
C. encouraging D. harmonious

2 Choose the best answer for each question.

(1) Currently, where is the automobile launched into space by Elon Musk?

A. It did not enter orbit and is still traveling through space.

B. It has returned safely to Earth.

C. It is on the surface of Mars.

D. It is orbiting Mars.

(2) How does the author feel about the use of robots in space exploration?

A. Robot technology will never be sufficient to replace humans.

B. Robots could carry bacteria harmful to the Martian environment.

C. Robots lack the skills necessary for space travel.

D. Using robots would be more economical and safer than humans.

(3) What do most people in the US believe NASA's principle mission should be?

A. developing robot technology

B. establishing a colony on Mars

C. preserving the environment on Mars

D. solving problems on Earth such as climate change

(4) What reason does the author NOT give against humans exploring Mars?

A. Humans would be bored on the trip.

B. Space is a dangerous environment for humans.

C. The main motivation for Mars exploration by humans is national pride.

D. The resources now used preparing for human space travel could be used to solve more urgent problems on Earth.

(5) Which best summarizes the author's opinion about human exploration of Mars?

A. A human colony on Mars would be a historic technological achievement.

B. It is dangerous, expensive, and inefficient.

C. It is necessary to solve the climate crisis.

D. Mars is a wasteland of no interest to humans.

Ⅳ

Answer the questions below.

1 James is very proud of his garden where he spends many enjoyable hours working every year. The weeding is hard work, but one of the rewards is the beautiful display of flowers that appears in spring each year. One day, he carefully picked 12 flowers. He used some of the flowers to make a bouquet for his grandmother's birthday party to celebrate her turning 80 years of age. Each flower in the bouquet represented 20 years of her life. Each flower was a different color, and he added in some green leaves to make the colors stand out more. He then chose some flowers to create a wedding anniversary gift for his wife. They had been married 12 years, and each flower he gave her represented three years of marriage. He also gave one flower to his neighbor and then arranged the remainder on his office desk.

How many flowers did he have on his office desk?

2 Mary is considering attending a university in California, and she is trying to calculate the costs with her parents. The price of university has risen considerably over the last few years, so she wants to make sure that it is not a big financial burden on her parents. She needs to determine how much she is going to need to borrow in total through a student loan. The cost of tuition for the university course is $5,000 per year. The course lasts four years. There are additional processing fees at the beginning of the first year that cost half as much as one year's tuition. She estimates that textbooks and other school supplies will cost $1,000 per year. Mary knows that she can get government assistance of $800 per year. Her parents have said that they can afford to give her $10,000 in total as a gift. In addition, she plans to work part-time and expects to earn $2,000 per year. She can live with a relative in California, but her living expenses will still be $2,500 per year.

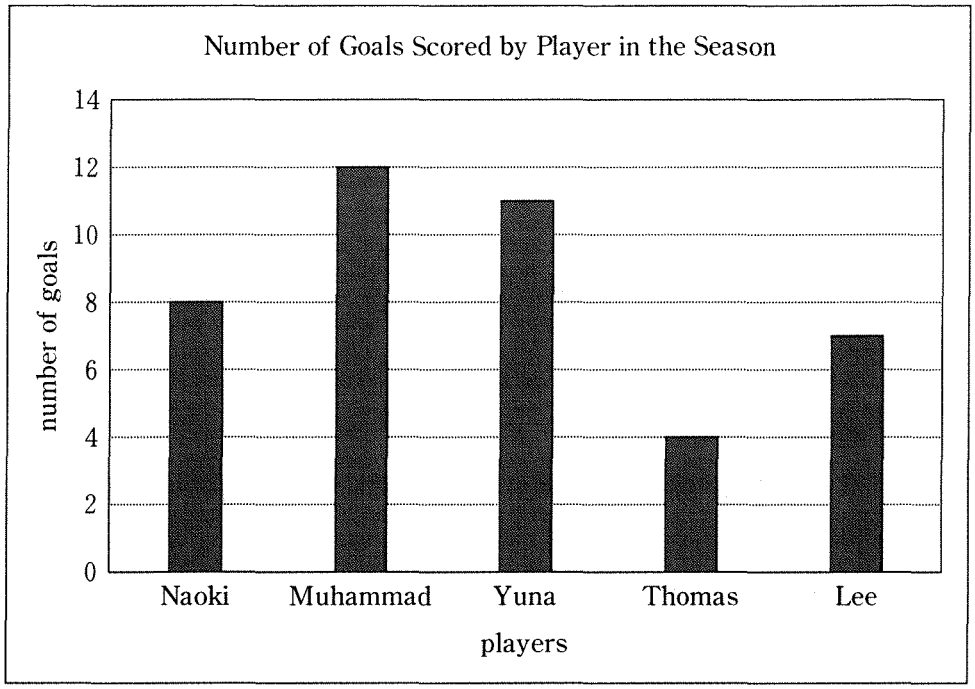
Mary plans to study at the university for four years. According to her plan, how much money will she need to borrow in total through the student loan?

3 Four friends — Emma, Finn, Grace, and Henry — are participating in a game. Each of them is assigned a number from one to four. Each person has a different number. The winner of the game is the person who can most quickly identify everyone's number based on the following clues.

Emma's number is half of Grace's number. Finn's number is one more than Grace's number. Henry's number is one more than Finn's number.

What number is each friend assigned?

4 Choose **the statement or statements** from A—F to describe the graph correctly. Circle the letter(s) on the answer sheet.



A. Muhammad scored fewer goals than Thomas and Lee combined.

B. Muhammad scored the most goals while Lee scored the fewest.

C. One-sixth of the five players' goals were scored by Lee.

D. Over half of all the five players' goals were scored by Muhammad.

E. The combined number of goals scored by Lee and Naoki is greater than the combined number of goals scored by Muhammad and Yuna.

F. Thomas scored less than half the goals that were scored by Yuna.

V

以下の問いに答えなさい。

食に関する地産地消とは，その地域で採れた食べ物をその地域で消費することをいいます。地産地消にはメリットがある一方で，課題も残っています。以下の表は，そのいくつかを記述したものです。

メリットの例

- 地元の農家や地元企業の経済活動を支えることで，その地域の経済が活気づく。
- 食品の産地が明確となり，農薬の使用の有無など安全性や品質に関して情報が得やすい。
- 食品や製品の輸送距離が短いため，輸送に伴う二酸化炭素の排出量が減る。
- 地域ごとの伝統的な農産物や特産品を支え，地域固有の文化や風土が守られる。
- 地域の人々のつながりを強化する。

課題の例

- 地域が限定されるため，天候や季節の影響を受けやすく，一定の供給量を確保しにくいことがある。
- 大量に作ることが難しいため，輸入品よりも高くなることがある。
- 地元で作られる食べ物だけだと手に入る種類が限られ，食の多様性が減少する。
- インフラが十分でない地域の場合，流通や保存にかかるコストが高くなる。

このような状況をふまえて，地産地消をさらに推進するアイデアを 60 語から 80 語の英語で具体的に書きなさい。